Assignment -1 (Innovative Assignment)

Institute Name:	Silver Oak College of Computer Application	**Programme Name:**	Master of Science Data Science
Semester:	1	**Academic Year:**	2026-2027
Course Name:	Information and Data Security	**Course Code:**	DSM5061C

Integrated Information and Data Security Investigation: Cyber Threats, Privacy, Security Technologies, Data Governance and Responsible AI .

Integrated Information and Data Security Investigation Portfolio: Real-World Data Breaches, Cybercrime, Privacy, OSINT, DPDP Act 2023, Security Technologies, IPR and Responsible AI

Students will select one real-world data breach, cyber incident, digital platform, or data-driven organization and conduct an integrated investigation covering information security, data security, privacy, cybersecurity technologies, cybercrime awareness, OSINT, data governance, the DPDP Act 2023, ethical data usage, Responsible AI, and Intellectual Property Rights.

Students must analyze the selected case using authentic and publicly available information and develop a practical security, privacy, governance and risk-mitigation solution.

The investigation should demonstrate the complete relationship:

Threat → Vulnerability → Data/Security Incident → Impact → Privacy Risk → Security Controls → Cybercrime Response → OSINT → Data Governance → DPDP Act 2023 → Ethics → Responsible AI → IPR → Risk Mitigation.

Assignment Submission Guidelines:

Particular	Guidelines
Assignment Type	Innovative Integrated Investigation
Submission Mode	Soft Copy (PDF) through Google Classroom
Deliverables	Project Report (20–30 Pages Recommended), Analytical/Creative Outputs, Security Framework, and Presentation (10–15 Slides)
Individual/Group	Individual / Faculty-approved Group
Total Marks	100
Total Workload	40 Hours
Report Format	A4 Size, Portrait, Times New Roman, 12 pt, 1.5 Line Spacing
Citation Style	IEEE / APA
File Naming Convention	`EnrollmentNo_StudentName_InformationDataSecurity.pdf`

Suitable Titles

Students may select an appropriate title from the following:

1. Integrated Information and Data Security Investigation of a Real-World Cyber Incident
2. Data Breach Investigation and Security Risk Assessment of a Digital Platform
3. Cyber Threat, Privacy and Data Security Analysis of a Real-World Incident
4. Information and Data Security Investigation: From Cyber Threats to Risk Mitigation
5. Integrated Cybersecurity and Data Privacy Investigation using Real-World Cases
6. Data Security, Privacy and Governance Analysis of a Data-Driven Organization
7. Cyber Threat Intelligence, Data Privacy and Security Technology Investigation
8. Information Security and Data Protection Analysis using Real-World Cyber Incidents
9. Data Breach, Privacy, Governance and Responsible AI Investigation
10. Integrated Information and Data Security Portfolio: Cyber Threats, Privacy, Governance and Responsible AI

Task

select **ONE real-world cyber incident, data breach, digital platform, or data-driven organization** and conduct a comprehensive investigation covering the following areas:

1. Case Study and Data Breach Analysis

Students must:

- Identify the selected organization/case.
- Explain the background and nature of the incident.
- Identify the type of cyber threat.
- Identify vulnerabilities and attack vectors.
- Analyze the data affected.
- Examine the impact on individuals and organizations.
- Identify the root cause.
- Study incident response and mitigation measures.

2. Privacy and User Consent Analysis

Students must analyze:

- Data collection practices.
- Types of personal data collected.
- Purpose of data collection.
- Data processing.
- Data sharing.
- Privacy policy.
- User consent mechanisms.

- Data retention.
- User control.
- Privacy risks.

Students should refer to the **official privacy policy** of the selected organization wherever available.

3. Security Technology Analysis

Students must examine the role and applicability of:

- Encryption
- Password hashing
- Authentication
- Multi-factor authentication
- Access control
- Role-Based Access Control
- Least privilege
- Cloud security
- Identity and Access Management
- Security monitoring
- Logging
- Backup and recovery
- AI-assisted security
- Anomaly detection
- Fraud detection
- Intrusion detection

Students should identify which security technologies could have prevented or reduced the impact of the selected incident.

4. I4C-Based Cybercrime Analysis

Students must use the official **Indian Cyber Crime Coordination Centre (I4C)** resources:

I4C – Indian Cyber Crime Coordination Centre

Students should:

- Identify the relevant cybercrime category.
- Analyze the cybercrime scenario.
- Study cybercrime prevention measures.
- Examine victim-response mechanisms.
- Study cybercrime reporting awareness.
- Identify appropriate evidence-preservation practices.
- Explain the relevance of coordinated cybercrime response.

5. Ethical OSINT Investigation

Students must use the **OSINT Framework**:

OSINT Framework

Students should collect and analyze **lawfully available public information** related to the selected case.

Possible areas include:

- Public organization information
- Public domain information
- Public technology information
- Public security disclosures
- Public incident reports
- Public news
- Public social-media information
- Publicly documented technology footprint

Students must provide an **OSINT Evidence Table** containing:

Evidence	Public Source	Relevance	Reliability

Students must not perform unauthorized access, exploitation, credential collection, intrusive scanning, or any activity against real systems.

6. Data Governance Analysis

Students must analyze:

- Data ownership
- Data stewardship
- Data classification
- Data collection
- Data quality
- Data storage
- Data access
- Data sharing
- Data retention
- Data security
- Data lifecycle
- Accountability

Students should identify **data governance gaps** and recommend appropriate controls.

7. DPDP Act 2023 Analysis

Students must analyze the selected case in relation to relevant concepts of the **Digital Personal Data Protection Act, 2023**, including:

- Personal data
- Data processing
- Notice
- Consent
- Purpose of processing
- Data protection responsibilities
- Security safeguards
- Personal data breach
- User rights
- Data retention
- Accountability

Students should prepare a **DPDP Analysis/Mapping Table**:

Area	Existing Practice	Risk/Gap	Recommended Measure
Personal Data			
Consent			
Notice			
Processing			
Security			
Data Breach			
Retention			
User Rights			

Students must distinguish between **legal requirements, organizational practices, observations and recommendations**.

8. Ethical Data Usage and Responsible AI

Students must analyze ethical issues related to:

- Privacy

- Fairness
- Transparency
- Accountability
- Bias
- Data minimization
- Responsible data collection
- Data misuse
- Human oversight
- Explainability
- Responsible AI deployment

Students should prepare a **Responsible AI Risk Matrix** wherever AI is applicable.

9. Intellectual Property Rights Analysis

Students must use the official **IP India** portal:

IP India – Intellectual Property India

Students should investigate relevant:

- Patents
- Copyright
- Trademarks
- Industrial designs
- Trade secrets
- Software/IP ownership
- Digital content
- Technology innovation
- AI-related intellectual property

Students must identify at least **one relevant IPR issue** related to the selected organization, product, software, technology or digital content.

10. Integrated Risk Assessment

Students must prepare a risk assessment covering:

- Data security
- Privacy
- Access control
- Cloud security
- Cybercrime
- Governance
- AI
- IPR

The risk should be classified as:

Low / Medium / High / Critical

11. Proposed Security and Governance Solution

Students must develop an integrated framework showing:

Data Collection → Privacy & Consent → Data Classification → Encryption → Access Control → Cloud Security → Monitoring → AI-Assisted Security → Incident Response → Cybercrime Response → Data Governance → DPDP Compliance → Ethical Review → Responsible AI → IPR Protection

The framework may be presented as:

- Architecture diagram
- Flowchart
- Infographic
- Security framework
- Digital dashboard
- Website prototype
- Mobile application prototype
- Cybersecurity toolkit
- Awareness solution

Suggested Innovative Solutions

Students may develop one or more of the following:

- Website Prototype
- Mobile Application Prototype
- Cybersecurity Awareness Portal
- AI Chatbot for Data Security Awareness
- Privacy Awareness Toolkit
- Cybercrime Awareness Campaign
- OSINT Investigation Dashboard
- Data Breach Risk Assessment Tool
- Privacy Risk Assessment Framework
- DPDP Compliance Checklist/Toolkit
- Data Governance Framework
- Responsible AI Framework
- Security Architecture
- Cybersecurity Infographic Series
- Educational Video
- Digital Handbook
- Cybersecurity Awareness Poster
- Incident Response Framework
- Cybersecurity Policy Framework

Required Original Outputs

Each student/group must prepare at least **THREE original analytical outputs**.

Students may select from:

1. Attack Flow Diagram
2. Vulnerability Map
3. OSINT Evidence Map
4. Privacy Gap Analysis
5. DPDP Compliance Matrix
6. Risk Assessment Matrix
7. Risk Heat Map
8. Security Architecture
9. Responsible AI Risk Matrix
10. IPR Mapping
11. Incident Response Workflow
12. Data Lifecycle Diagram

Report Structure

The submitted report should contain the following sections:

Section	Required Content
1. Introduction	Background and importance of Information and Data Security
2. Case/Organization Background	Organization, technology, users and selected incident
3. Problem Statement	Clearly defined security/privacy problem
4. Threat Analysis	Threats, attack vectors and threat actors
5. Vulnerability Analysis	Security weaknesses and exploitation
6. Impact Analysis	Technical, financial, privacy and societal impact
7. Privacy & Consent Analysis	Privacy policy, data collection and user consent
8. Security Technology Analysis	Encryption, access control, cloud and AI security
9. I4C Analysis	Cybercrime awareness, reporting and response
10. OSINT Investigation	Public information and evidence analysis
11. Data Governance	Data lifecycle, access, storage, sharing and accountability
12. DPDP Act 2023	Data protection and compliance analysis
13. Ethical Data Usage	Ethical risks and responsible practices
14. Responsible AI	AI risks, fairness, transparency and human oversight
15. IPR Analysis	Patent, copyright, trademark, trade secret etc.
16. Risk Assessment	Risk identification and prioritization
17. Proposed Solution	Integrated security and governance framework
18. Recommendations	Practical mitigation and improvement measures

19. Conclusion	Key findings and learning
20. References	IEEE/APA references
21. Appendix	Screenshots, evidence, diagrams and supporting material

Recommended Report Format

Particular	Guidelines
Page Size	A4
Orientation	Portrait
Font	Times New Roman
Font Size	12 pt
Heading	14–16 pt Bold
Line Spacing	1.5
Alignment	Justified
Report Length	20–30 Pages Recommended
Citation	IEEE / APA
Figures	Number and provide captions
Tables	Number and provide captions
Screenshots	Provide source/reference
Submission	PDF
File Name	`EnrollmentNo_StudentName_InformationDataSecurity.pdf`

Presentation Guidelines

Students must prepare a **10–15 slide presentation**.

Suggested Slide Structure

1. Title
2. Case/Organization Background
3. Problem Statement
4. Threat and Vulnerability Analysis
5. Data Breach and Impact
6. Privacy and Consent
7. Security Technologies
8. I4C and Cybercrime Perspective
9. OSINT Investigation
10. Data Governance and DPDP Act 2023
11. Ethical Data Usage and Responsible AI
12. IPR Analysis

13. Risk Assessment
14. Proposed Solution
15. Conclusion and Recommendations

Students must be prepared for **viva questions** based on their investigation.

Mandatory Resources

Resource	Purpose
I4C – Indian Cyber Crime Coordination Centre	Cybercrime awareness, prevention, reporting and investigation-related study
OSINT Framework	Ethical open-source information investigation
IP India	Patents, trademarks, copyright, designs and IPR investigation
Official Organization Website	Organizational information
Official Privacy Policy	Privacy and consent analysis
Government Sources	Legal and regulatory information
Research Papers	Academic evidence
Reputed Cybersecurity Reports	Incident and threat information

Academic Integrity and Ethical Guidelines

Students must:

- Use authentic and reliable sources.
- Cite all external information.
- Avoid plagiarism.
- Use original analysis.
- Clearly distinguish facts from personal interpretation.
- Use lawful OSINT techniques.
- Never attempt unauthorized access.
- Never obtain passwords or credentials.
- Never exploit a real website/system.
- Never conduct unauthorized vulnerability testing.
- Avoid collecting unnecessary personal information.
- Do not fabricate evidence.
- Do not make unsupported legal claims.
- Properly acknowledge all images, screenshots, datasets and external materials.

Innovative Assignment Rubrics for Evaluation

Evaluation Criteria	Description	Marks
Problem Identification & Case Selection	Selection of a relevant real-world data security/cyber incident, clarity of the problem statement, case background and justification for selection.	10
Threat, Vulnerability & Impact Analysis	Identification and analysis of threats, vulnerabilities, attack vectors, affected data, root causes and technical, financial, privacy and societal impacts.	15
Privacy & User Consent Analysis	Evaluation of data collection, processing, sharing, privacy policies, consent mechanisms, user control and privacy risks.	10
Security Technology Analysis	Analysis and comparison of encryption, authentication, access control, cloud security, monitoring and AI-assisted security solutions.	10
I4C & Cybercrime Analysis	Application of I4C resources to cybercrime awareness, prevention, reporting, victim response, evidence preservation and cybercrime coordination.	10
Ethical OSINT Investigation	Appropriate use of publicly available information, evidence collection, source reliability, ethical practices and security relevance.	10
Data Governance & DPDP Act 2023	Analysis of data governance practices, personal-data protection, consent, security safeguards, breach management, user rights and relevant DPDP concepts.	10
Ethical Data Usage & Responsible AI	Evaluation of privacy, fairness, transparency, accountability, bias, explainability, human oversight and responsible AI practices.	5
IPR Analysis using IP India	Identification and analysis of relevant patents, copyright, trademarks, designs, trade secrets or software/IP issues using authentic IP India resources.	5
Innovation & Proposed Solution	Creativity, originality, feasibility and effectiveness of the proposed security, privacy, governance, awareness, website/app/toolkit or framework solution.	5
Risk Assessment & Mitigation	Quality of risk identification, prioritization and practical mitigation recommendations.	5
Report Documentation & References	Organization, methodology, formatting, diagrams, tables, evidence, references, citation quality and overall documentation.	5
Presentation & Viva	Quality of presentation, communication, technical understanding, demonstration, confidence and responses to questions.	5
Total	**Overall Evaluation**	**100**

Course Coordinator **HoD / HoI**

SILVER OAK UNIVERSITY
EDUCATION TO INNOVATION

Assignment - 2 (Handwritten Assignment)

Institute Name:	Silver Oak College of Computer Application	**Programme Name:**	Master of Science Data Science
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Q. No.	Assignment Question
Q1	Analyze a real-world data breach and identify the information assets, threats, vulnerabilities, risks, CIA Triad violations, authentication weaknesses, and cryptographic controls involved.
Q2	Design a data privacy lifecycle for a digital application and identify privacy risks and protection techniques at each stage of data collection, storage, processing, sharing, and disposal.
Q3	Develop a security strategy for a cloud-based AI/data-driven system by identifying network, cloud, big-data, AI, adversarial attack, data poisoning, deepfake, and Generative AI security risks.
Q4	Develop a data governance and compliance framework for an organization handling personal data, incorporating DPDP Act 2023, ethical AI, security policies, ISO 27001, and NIST Cybersecurity Framework principles.

Rubrics for Evaluation

Criteria	Marks
Concept Understanding	5
Analysis & Application	5
Solution / Recommendations	5
Presentation & Innovation	5
Total	**20**

Course Coordinator **HoD / HoI**